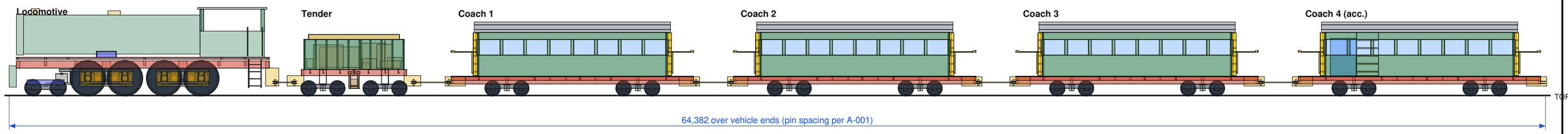
# PULSIPHER FARMS CHRISTMAS TRAIN — DESIGN SITE DRAWING SET PF-CT REV L4.2

36 in gauge diesel-hydraulic locomotive + tender + 4 coaches · 6 mph GOVERNED (42 cc stroke stop + sealed 2,600 rpm; hardware 7.6 mph; trip 7 mph) · 100 m min. radius · track LEVEL (Layout F 0.0 %), vehicles checked at 2 % (conservative)




## GENERAL NOTES

1. PRELIMINARY. Owner engineering for review by a Nevada-registered PE; not for fabrication until stamped and released.
2. Codes (to verify by PE): Clark County Code Ch. 22.16 / Ord. 5378; ASTM F2291, F1193, F770, F3598 (amusement rides); AISC 360 (steel), AWS D1.1 (steel welding) / D1.3 (sheet), IBC 1607 for guard, rail and stair loads.
3. Materials: HSS ASTM A500 Gr C (Fy 50 ksi); plate & bar ASTM A36 unless noted; drawgear plate A572 Gr 50; pins and axles AISI 4140 Q&T 28-32 HRC (Fy >= 95 ksi, mill cert required). Mill certificates for all primary steel.
4. Welding: GMAW/FCAW E70 to AWS D1.1 by qualified welders under a written WPS. Default fillet = 6 mm all round unless sized on sheet. Primary drawgear welds (PF-DB boxes): 10 mm fillet all round, 100% visual + MT.
5. Bolts: structural ASTM F3125 A325 / SAE J429 Gr 8 / ISO 898 cl. 8.8-10.9 as noted; hardened washers; prevailing-torque nuts or Nord-Lock on running gear. Torque per fastener schedule, mark after torque.
6. Tolerances: ISO 2768-mK unless noted; cut lengths ±1.5, holes ±0.8, frame diagonals within 3 mm, pin bores H8. GD&T per ASME Y14.5-2018 where shown (L-103.2 final drive, axle seat); weld symbols per AWS A2.4 (schedule S-503).
7. Coupling height 550 mm ARL on all vehicles (empty, new wheels). Max wheel load basis 26 kN (K2-a 25.89 kN, L4.2). GOVERNED 6 mph max (owner-selected, PE to confirm).
8. Items marked AWAIT are waiting for supplier data (AIE/Kubota, Poclain, chain/sprocket maker, Danfoss/PLUS+1, IR, ZF/WABCO). Do not invent values.
9. Third-angle projection. Fastener torques per S-503 (0.75 x proof load; K 0.20 dry / 0.14 lubricated).
10. Calculations: PF-CT_PE_calc_package.pdf (rev L4.2) + calcs/L4/drive_L4.json. Open items / assumptions: PF-CT_assumptions_open_items.pdf.
11. Sheet numbers follow the Loco Bot shop-sheet master (L-1xx): L-103 final drive/motor mounting (+ L-103.2), L-105 hydraulic routing, L-106 front ballast; shop sheets L-103, L-105, L-106, L-107L, L-108, L-109, L-110, L-111, L-114, S-504, S-507, A-101 form part of this set (index). Track basis: Layout F LEVEL 0.0 %; vehicles checked at 2 %.

## REVISION HISTORY

| Rev | Date | Description |
|---|---|---|
| R1 | 2026-10-27 | Initial issue for owner / PE review (12 sheets). |
| R2 | 2026-10-27 | Owner decisions: emergency stairs all coach doors (C-304, F08); |
| | | 40 mph gust no-run + tie-downs DECIDED; roof 3,700 ADOPTED, |
| | | cab headroom ~1,960, route overhead min. 4,300 ARL; pony open (rec. delete). |
| L1 | 2026-10-27 | Owner: loco wheels 'like the plans'. Running gear per LT18-LT20: |
| | | 8x Ø1,450 drivers, 1,550 wb, inside frames, leading truck Ø650 (worn limit Ø630), unbraked, running |
| | | board Z1,605. 2% start now motor-short at 1:1 (OPEN F25, C13). |
| L1b | 2026-10-27 | Owner: no chains (F25 chain options dropped). Model corrected to 4 motors, |
| | | one per axle, staggered L/R per plan DT4 (was 8). Drive choice OPEN F25. |
| L2 | 2026-10-27 | Owner chose F25 (c): inboard spur-gear case 21:38 (1.81:1) per axle; MSE05 |
| | | inboard, nose link to transom, pinion-shaft disc brake, sanders. Outboard |
| | | DT5 cartridges, couplings, motors removed. F03, F25 RESOLVED (L-101, L-103). |
| L2 | 2026-10-27 | PE-review pass: title blocks w/ tolerance + PE stamp box; added L-106, S-501..503, |
| | | B-601, E-701; rail basis 26 kN; chevron springs; HSS3x2 roof bows; ERVs. |
| L2s | 2026-10-27 | Supplier fold-in: H1P053 G1 shaft + anti-stall, L35 350/400 bar, charge <=30 bar, |
| | | case drains, spline motor shaft, eta 0.92/0.78, SKF gear-case brgs, Coremo ref.; |
| | | ERV RenewAire HE05IN roof pod, 62.1 OA (E-701). |
| L3 | 2026-10-28 | Owner: best fit. Catalog Dodge TA7315H15MS 14.9:1 + Danfoss H1B110 via 1:1 |
| | | belt drop (L2 custom case rejected); Ø120 seat 34CrNiMo6 ind.-hardened; axle |
| | | disc + 2 calipers; TA rod to transom. GOVERNED 6 mph (owner, PE to confirm). |
| | | L-101, L-103, L-106 (replaced), S-501, S-503, B-601, G-001. |
| L4 | 2026-10-28 | Owner: no belt. Poclain MS05 468/234 2-speed + T04 brake, frame-mounted; |
| | | ANSI 120-2 15T:45T 3:1 chain in oil bath; Ø150 pressed hub, no key; (flow divider |
| | | dropped L4.1); stroke stop 6 mph; independent 7 mph trip; 650 kg front |
| | | ballast + 1.25 t sprung pony; cyl. Y±790; IR 2340 + 14 kW cooler on tender. |
| | | L-101, L-103, L-106 (replaced), T-201, K-401, S-501, S-503, B-601, G-001. |
| L4.1 | 2026-10-28 | RELEASED. NO flow divider: PLUS+1 traction control (pony ref., destroke, auto-sand). |
| | | Shaft code 5 W55x3x17; 8.8 fitted ring bolts 219 N.m; hub SHRINK 0.15-0.20; low traction |
| | | pin h 0.65; trip ramp 3 s + 2nd default + T04 switch/choke; bump stops ±22; 38 cc stop; |
| | | ballast 2 × 75.4 + clamp bars. L-101, L-103, L-105, L-106, S-501, S-503, B-601, G-001. |
| | | s7: T04 release from loop (LP shuttle, 20 bar PRV, NV solenoid, choke + 1 L acc.; manual 6 L |
| | | acc. + hand pump), no 4th dry-break; condemn drivers Ø1430, flange ≤ 30 mm; reverse ≤ 1.55 %. |
| | | s8/9: tender TB-3.9 8,616.3 kg (0.676 m/s²); loco release ≤ 4 s (C ≥ 3.0, 3/4 in trunk, 1/2 in relay); |
| | | safety-chain lugs PL40 CJP + PL16 diaphragms (+10.2 kg). B-601, S-503, S-501. |
| L4.2 | 2026-10-28 | Stress Bot r3 rulings: FLOAT descents (bypass, >= 3 s ramp); T04 from charge, Ø1.2 orifices, 2-ch switch; |
| | | 42 cc + sealed 2,600 rpm; flange 28 / re-profile <= 27; traction pin Ø120 + GE120ES; sanders both trucks; |
| | | reverse start <= 1.20 % (28.9 kN cap); HPRV 380, loop parts >= 420 WP, strength 430; chain cert >= 326 kN; |
| | | 4140 15T; NB-1 59.8 kg; lugs PL44.5; tender TB-3.13d; loop hose 560 bar; tow valve DN06 500; renumbered L-103.2, L-106. |

## SHEET INDEX

| Sheet | Title |
|---|---|
| G-001 | Cover, general notes and sheet index |
| A-001 | Train general arrangement |
| L-101 | Locomotive general arrangement |
| L-102 | Locomotive frame and member schedule |
| L-103 | Powered truck + final drive / motor mounting (Ø1,450, MS05 + 120-2 chain) |
| L-103.2 | Final drive unit - MS05 + 120-2 15T:45T, 4140 hub, GD&T (was L-106) |
| L-104 | PF-DB drop drawbar box, drawbar, pin, chains |
| L-106 | Front ballast pockets (P-106), cab steps and grab irons (was L-105) |
| T-201 | Tender frame and general arrangement |
| C-301 | Coach general arrangement |
| C-302 | Coach underframe and member schedule |
| C-303 | Coach truck |
| C-304 | Coach emergency / boarding stairs (R2) |
| K-401 | Track, rail and clearance outline |
| S-501 | Bill of materials - locomotive |
| S-502 | Bill of materials - tender T2, coach R2, drawgear |
| S-503 | Weld and fastener schedule |
| B-601 | Brake, air and hydraulic schematic |
| E-701 | Electrical one-line and coach HVAC |
| | LOCO BOT SHOP SHEETS (master numbering; part of this set): |
| L-103 shop | Final drive + motor mounting shop detail (P-103-1 hanger) |
| L-105 shop | Loco hydraulic routing (loop parts >= 420 bar WP) |
| L-106 shop | Front ballast (P-106-1 clamp bars, P-106-2 pockets) |
| L-107L | Running-board supports, light option (28 HB16) |
| L-108 | Boiler barrel shell and rings |
| L-109 | Smokebox, front and pilot (NB-1 nose ballast) |
| L-110 | Domes, stack and fittings |
| L-111 | Cab structure C1 + FT-1 tow point |
| L-114 | Loco electrical/pneumatic interfaces |
| S-504 | Loco body BOM + cut lists |
| S-507 | Loco weld takeoff |
| A-101 | Loco build and assembly sequence (weigh-in hold point) |

**PE STAMP / SIGNATURE**

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**UNLESS NOTED: ISO 2768-mK general tolerances**
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

**PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)**

**Cover, general notes and sheet index**

Sheet G-001 · REV L4.2 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

**PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED**